AI101 MINI-PROJECT · GROUP 73

PREDICTING GLOBAL ENERGY CONSUMPTION

Estimating a country’s yearly primary energy use (TWh) from its population, GDP and year with linear regression.

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Repo: github.com/Aadyant-Chaturvedi/Predicting-Global-Energy-Consumption

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Problem AND DATA

WHAT AND WHY

Predict a country's yearly primary energy consumption (TWh) from its population, GDP and year.

A simple, explainable baseline for demand forecasting.

CLEANING

ISO-coded countries only (regions double-count energy), 1990 onward, rows missing target, population or GDP dropped.

22,012
rows in raw file

129
columns in raw file

4,732
rows used after cleaning

165
countries used (of 219)

1990–2018
years used

3
input features

Source: Kaggle · World Energy Consumption (pralabhpoudel) · CC BY 4.0

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Approach

STEP 1

Data

- World Energy Consumption (Kaggle)
- 22,012 rows × 129 columns
- Raw file covers 1900–2022

STEP 2

Preprocess

- ISO-coded countries only
- Years 1990 onward
- Drop rows with missing values
- X: population, GDP, year
- y: primary energy use (TWh)

STEP 3

Model

- Random 80/20 train/test split (seed 42)
- LinearRegression from scikit-learn
- Fit on training rows, predict on test rows

STEP 4

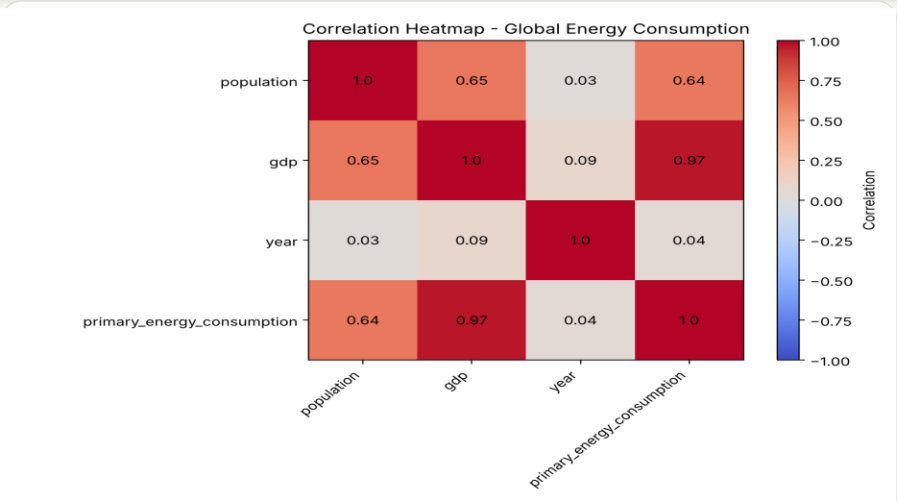
Output

- R^2, MAE and RMSE printed
- Predicted vs actual plot
- Correlation heatmap
- Predicted and test CSVs

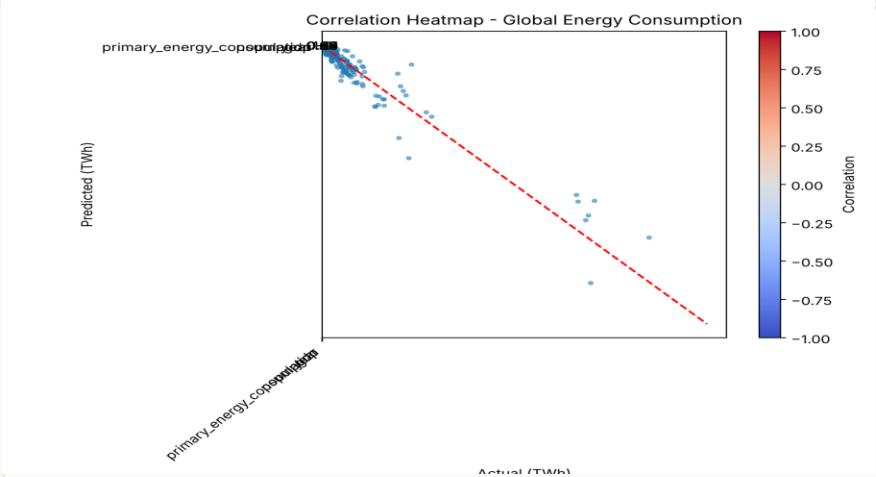
Libraries used pandas NumPy scikit-learn Matplotlib

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Results



Correlation heatmap: GDP r = 0.97



Predicted vs actual, test set

R² SCORE
0.93

MAE
248.9 TWh

RMSE
683.1 TWh

Demo: terminal run

```
$ python main.py
```

```
R2:  0.93
MAE:  248.9 TWh
RMSE:  683.1 TWh
```


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Learnings

What worked

- GDP tracks energy use closely (r = 0.97)
- Three inputs and a linear model reach R^2 = 0.93
- The whole pipeline runs in about 5 seconds

What didn't

- A few very large countries dominate the RMSE; North Korea is the biggest miss (data largely misreported)
- Only 165 of 219 countries covered: rows missing GDP or population are dropped
- Year adds almost nothing (r = 0.04)

Who did what

- Aadyant Chaturvedi: Documentation Lead
- Ayush Yogesh Sakure: Execution Lead
- Padakandla Rohith: Video Lead
- Sahil Shekhawat: PPT Lead
- All members contributed code

AI-use disclosure

Claude was used to draft the outline of the README and to run preliminary data analysis on which variables would add the most value to the model. Claude also helped build these slides.

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